5.15

ASSIGNMENT : # 8

The total energy of a free electron system with a total of Nq electrons is →

$$E_{Tot} = \frac{\hbar^2 [3\pi^2 N_q]^{5/3} V^{-2/3}}{10\pi^2 m} \rightarrow [5.45]$$

The total no. of electrons & the Fermi wave number are related →

$$k_F = \left[3 \frac{Nq}{V} \pi^2\right]^{1/3}$$

Thus the Fermi energy is →

$$E_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{3Nq\pi^2}{V}\right)^{2/3}$$

$$\Rightarrow E_{Tot} = \left(\frac{3}{5} E_F\right) Nq$$

$$\Rightarrow \frac{E_{Tot}}{Nq} = \frac{3}{5} E_F$$

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(a) Fermi Energy →

$$E_F = \frac{\hbar^2}{2m_e}\left[3\rho\pi^2\right]^{2/3}$$

$$\rho = \frac{Nq}{V} \; ; \; \boxed{\text{Assuming } q = 1}$$

$$= \frac{N}{V} \equiv \text{No. of copper atoms / unit volume}$$

$63.5\,gm \longrightarrow 6.02\times10^{23}$ atoms

$8.96\,gm$ in $1\,cm^3 \longrightarrow 6.02\times10^{23}\times\frac{8.96}{63.5}$ atoms in $1\,cm^3$

$$\Rightarrow \rho = 8.49\times10^{28}/m^3$$

$$E_F = \frac{\dfrac{\left[1.05\times10^{-34}\,Js\right]^2}{2\times9.1\times10^{-31}\,kg}\left[3\pi^2\;8.49\times10^{28}\,m^{-3}\right]^{2/3}}{1.6\times10^{-19}\,C}$$

$$= 7.04\,eV$$

(b) $7.04\times1.6\times10^{-19}\,J = \frac{1}{2}\times9.1\times10^{-31}\,v^2$ $\left[\text{using } E_F = \frac{1}{2}mv^2\right]$

$$\Rightarrow v = 1.57\times10^6\,m/s$$

$$\Rightarrow \frac{v}{c} = 5.25\times10^{-3} \ll 1$$

Hence electrons are nonrelativistic.

(c) Characteristic thermal Energy →

$$K_B T = E_F$$

$$T = \frac{(7.04 \times 1.6 \times 10^{-19})\,J}{1.38 \times 10^{-23}\ J/K}$$

$$= 8.17 \times 10^{4}\ K$$

(d) Degeneracy Pressure

$$P = \frac{2}{3}\frac{E_{tot}}{V} = \frac{(3\pi^2)^{2/3}\hbar^2}{5m}\rho^{5/3}$$

$$= 3.84 \times 10^{10}\ N/m^2$$

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Bulk Modulus: $B = -V \dfrac{dP}{dV}$

$$P = \frac{(3\pi^2)^{2/3}\hbar^2}{5m}\underbrace{\left(\frac{Nq}{V}\right)}_{\rho}^{5/3}$$

$$\Rightarrow -V\frac{dP}{dV} = \frac{5}{3}V\left[\frac{(3\pi^2)^{2/3}\hbar^2}{5m}(Nq)^{5/3}\frac{1}{V^{5/3}V}\right]$$

$$= 5/3\, P$$

$$\Rightarrow B = \frac{5}{3}P = \frac{5}{3}\times 3.84\times 10^{10}\ N/m^2$$

$$= 6.4\times 10^{10}\ N/m^2 .$$

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$$f(z) = \cos(z) + \beta \frac{\sin(z)}{z}$$

$$\beta = 10$$

Graph of $f(z) \longrightarrow$ look at Fig 5.6

<u>Bottom</u> of the first allowed band corresponds to the solution of the equation : $1 = \cos z + 10 \frac{\sin z}{z}$,

where z lies between $0 < z < \pi$.

Solving for z one obtains : [using Maple/Mathematica]

$$z = ka = 2.62768$$

$$\Rightarrow E = \frac{\hbar^2}{2m} k^2 = \frac{\hbar^2}{2m} \frac{(2.62768)^2}{a^2}$$

$$\beta = 10 = \frac{m\alpha a}{\hbar^2} \Rightarrow \frac{\hbar^2}{2m} = \frac{\alpha a}{20}$$

$$\Rightarrow E = \frac{\alpha a}{20} \frac{(2.62768)^2}{a^2} = 0.345 \text{ eV}.$$

where $\alpha/a = 1\text{eV}$